MILITARY SPECIFICATION

RACK, TEST TUBE, LABORATORY, POLYPROPYLENE

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers autoclavable plastic test tube racks that are suitable for use in medical laboratories.

1.2 Classification. Racks covered by this specification shall be of the following capacities:

Size 1 - 40 test tube capacity to accommodate test tubes 16 to 20 mm. in diameter.

Size 2 - 72 test tube capacity to accommodate test tubes 10 to 13 mm. in diameter.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

FSC 6640

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Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

3. REQUIREMENTS

3.1 Material. The racks shall be fabricated from suitable polypropylene material that is capable of meeting the requirements of 3.3. If welding is used to bond components of the rack, the bonding material shall be the same material as the rack.

3.2 Design and construction. The racks shall be constructed in accordance with either Figure 1 (for size 1) or Figure 2 (for size 2), as applicable. Variations in contours of the rack from that indicated in the applicable figure are acceptable when not limited by requirements or dimensions. Each rack shall have a rectangular base of stiffened design and two upper tiers. The base and tiers shall be firmly secured to end support by means of either mechanical joints or welding. The upper and lower tiers shall have an equal number of properly aligned holes; and the base shall have the same number of countersunk holes to accommodate the test tube bottoms. The holes for the size 1 racks shall conform to the requirements of 3.2.1, and the holes for the size 2 racks shall conform to the requirements of 3.2.2. The center lines of corresponding holes in the base and tiers of the rack shall be in proper alignment to assure that the inserted tubes rest vertically in the rack. The cutout for each end support shall serve as a carrying handle. The top and bottom surfaces of the end supports shall be designed to permit secure stacking of the same size test tube racks. The underside of the rack shall have cutouts to accommodate projecting tabs from end supports of bottom racks during stacking.

3.2.1 Size 1 racks. The size 1 racks shall have 40 holes which are arranged in 4 rows with 10 holes per row equidistant from each other. Each hole shall be 7/8 inch in diameter with a tolerance of +0 and - 1/32 inch.

3.2.2 Size 2 racks. The size 2 racks shall have 72 holes which are arranged in 6 rows with 12 holes per row equidistant from each other. Each hole shall be 9/16 inch in diameter with a tolerance of +0 and -1/32 inch.

3.3 Performance. Each rack shall be capable of withstanding sterilization by autoclaving. When tested in accordance with 4.4.1, each rack shall not distort, disassemble, or become tacky. When tested in accordance with 4.4.2, the base and tiers shall remain secured to end supports without evidence of cracks or fractures to the plastic material.

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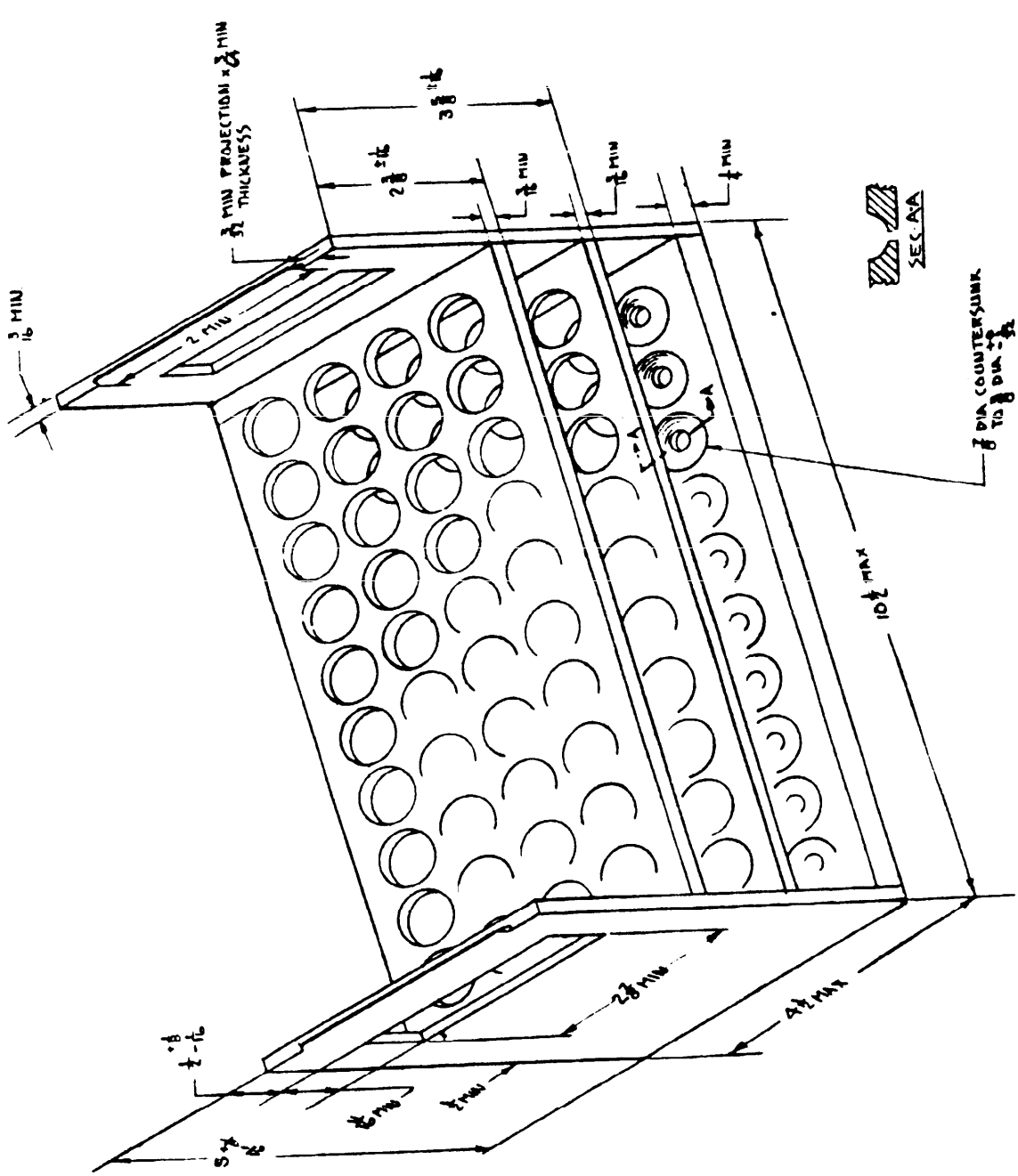


FIGURE 1

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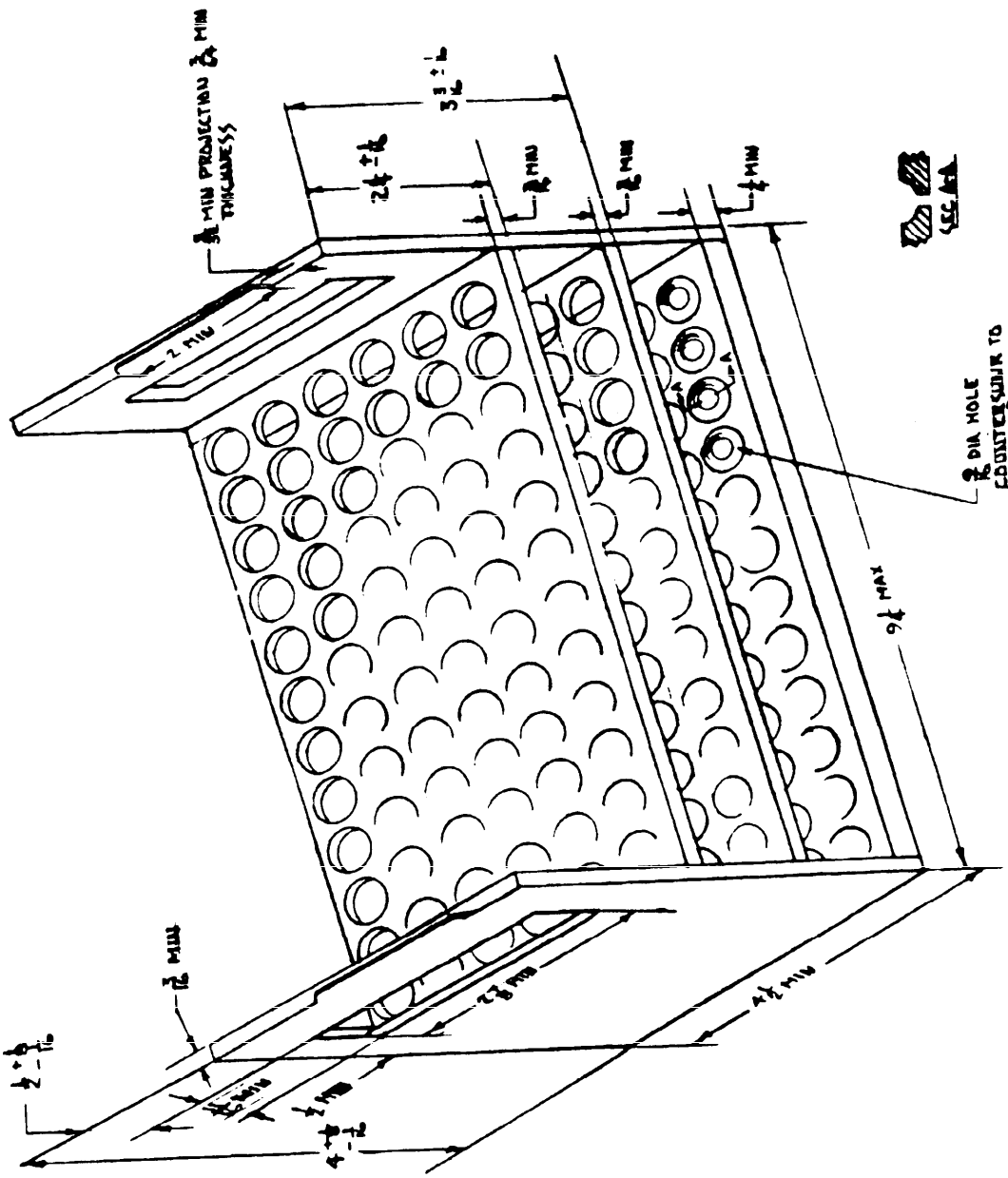


FIGURE 2

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3.4 Workmanship. Workmanship shall be first class throughout, and each rack shall be free from defects which detract from its appearance or may impair its serviceability. All surfaces (including cutouts) shall be smooth and free of sharp edges, surface craze, mold flash, or projections and burrs (except for mold marks due to gating). Mold lines shall not be cause for rejection provided there are no flash protrusions. Burrs due to gating shall not be cause for rejection, provided there is no evidence of sharpness or projections that protrude more than 1/64 inch from the surrounding area. Each rack shall not wobble when resting on a flat surface, or when stacked on top of another rack.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I. For Examination

	Inspection level	AQL (percent defective)
For visual examination:		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II. The acceptance number shall be zero.

Table II. For Tests

Characteristic	Requirement	Test procedure	Sample size
Autoclavability	3.3	4.4.1	*
Drop test	3.3	4.4.2	*

* Four racks per lot of polypropylene material.

4.3 Examination. The racks shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III. Examination shall not be restricted to the defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Major	
101	Rack not free from deformation or distortion.
102	Rack base and tiers not firmly secured to end supports.
103	Holes on shelves not properly aligned.
104	Rack not free of cracks, cuts, or depressions.
105	Rack and cutouts not free of sharp edges or sharp protrusions.
106	Rack not free of wobbling when resting on a flat surface.
107	Rack not free from embedded foreign particles.
108	End supports not properly constructed to allow for proper stacking of racks.
Minor	
201	Holes on upper or lower tier not round.
202	Rack and cutouts not free of rough surfaces (except for mold marks per 3.4).
203	Rack not free from dirt, grease, or removable foreign particles.
204	Rack not free of flash.
205	Rack not free of surface craze.

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4.3.2 Dimensional examination. The racks shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Autoclavability. The racks shall be subjected to steam sterilization in a suitable autoclave at 15 pounds per square inch (p.s.i.) for a period of 30 minutes. After the sterilization period, the racks shall be removed from the autoclave and examined for evidence of damage. The test on the racks shall be repeated two additional times.

4.4.2 Drop test. A sample rack shall be dropped on each corner of the base from a height of 36 inches to a concrete floor. The center of gravity of the rack shall be directly over the corner which strikes the floor.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One rack, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	4 units	16 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method III.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in the applicable table in PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of appicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. Special requirements as specified in the applicable table in PPP-B-636 shall apply for fiberboard boxes.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The racks are intended for holding test tubes for use in medical laboratories.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Size (see 1.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0699

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-36837 (DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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